15 **(a)** Show that, for the curve $y = ax^2 + bx + c$, the equation of the tangent at the point with x-coordinate t is $y = (2at + b)x - at^2 + c$. **[3]**

(b) Hence show that for the curve with equation $y = ax^2 + bx + c$, the tangents at two points, P and Q, on the curve cross at a point which has x-coordinate equal to the mean of the x-coordinates of points P and Q, as given in lines 11 to 14. **[3]**